



For industries, today, one needs flexible business models.

Also, without a comprehensive partner system, it is very difficult to cater to a complete IoT solution. One needs the right support from everyone, right from chipset manufacturers to cloud service providers. The support will help in development of a nice looking IoT solution that is market-ready.

Anatomy of a successful IoT product or solution is analogous to a healthy human body. We, as humans, try to survive for decades, and we strive to be healthy. Similarly, there is a need to work on the IoT systems that can survive not only for this decade but decades to come.

Hardware

In an IoT system, the heart of the ecosystem is hardware. One can easily get into a trap of developing hardware with multiple interfaces that have multiple applications. With this, the cost of hardware increases. Generic hardware with multiple applications, though good for pilot and PUCs, are not viable for a product targeting a specific use case. Once the application is decided, optimise the hardware with minimal interfaces so that the cost can be brought down. Once the cost is brought down, adoption increases and one can get into a circle of demand and supply.

It is necessary to think about implicit requirements. For example, an asset tracker with GPS might only take 2-3 months to develop from the software side. However, there is a high chance of failure if the following things are not taken care of: hardware watchdogs, device management, edge algorithms for the processing on cloud, firmware over-the-air (FOTA), reduction of cost, IP ratings to sustain in field environments, and power management.

The time to market will drasti-

cally reduce if all of these aspects are kept in mind from day one of development. These aspects can also be applicable in multiple genres of hardware to be created in the IoT sector.

Edge devices

Once the hardware has been developed, it is important to ensure that the data flows from and to the hardware, and the processed message flows to the cloud system. There can be various protocols for data transfer because of various proprietary technologies integrated. One must start working on protocols and manage libraries and SDKs for further reuse. The edge analytics on the device can process the data and send it to the cloud.

There are multiple cloud technologies available today. One of the most interesting from the perspective of a developer is NIDD, which is a part of NB-IoT (Narrowband IoT). This is the technology that you might look out for as it is going to revolutionise the way data transfers reliably from hardware to the cloud.

What has haunted us in this journey is weak indoor penetration, lower coverage area, vendor lock-ins, no FOTA support, high-power consumption, high infrastructure and maintenance cost, security vulnerabilities due to unlicensed spectrum, and shared resources among a large pool of devices. These challenges have blocked the arteries of the IoT system. The arteries can be unclogged with the help of open technologies for deep indoor penetration to reach the most difficult areas. For this low-power consumption, high security, coexistence with future technologies, no upfront infrastructure costs, high quality of services, low recurring costs, and non-IP data delivery to reduce payload overheads can help.

Technologies like SigFox need a lot of initial investment and are de-

pendent on people to set up their networks. Therefore, it is better to go with cellular technologies. For example, one of the most mature markets is VTS because people didn't have other options except 2G. Had there been something like LoRA, solutions would have been figured out around such technologies. Because there was no other option, VTS was adopted. Therefore, it is important to select the technology to adopt after due diligence and evaluation, and make sure it gets adopted by giving less non-viable options.

GDPR and device management

Device architecture is one of the most important aspects of an IoT solution, and many startups nowadays are proficient at these technologies. However, they miss out aspects like GDPR, device management, and FOTA, which are crucial. In addition, security is one of the primary aspects which must be taken care of from day one to succeed in the long run.

The architecture of the solution you build must support all these. There are many horizontal platforms that do these things for you. For startups, these may be available for free the first year, but could prove very expensive when they are actually ready to adopt the technology. Therefore, try to balance by building your own technology using the right open source applications available and the existing available technologies.

Installation and after-sale

These are the most critical aspects that are often neglected. Most of the startups and ventures focus on hardware or the platform, and the installation and after-sales services are left aside. One needs to lend ears to customer feedback and give good after-sale services.

Attention to each aspect in